

1 Assignment 1

Recall that given a formula $\varphi(X, Y)$ its *uniformisation* is a formula $\psi(X, Y)$ which satisfies the following conditions:

- $\forall X. \forall Y. \psi(X, Y) \Rightarrow \varphi(X, Y)$ — ψ is contained in φ ,
- $\forall X. (\exists Y. \varphi(X, Y)) \Rightarrow (\exists Y. \psi(X, Y))$ — the projections onto X agree,
- $\forall X. \forall Y_1, Y_2. (\psi(X, Y_1) \wedge \psi(X, Y_2)) \Rightarrow Y_1 = Y_2$ — ψ is functional.

Give an argument showing that the following problem is decidable: given $\varphi(X, Y)$ in MSO on ω -words over an alphabet A , decide if there exists an MSO formula $\psi(X, Y)$ which uniformises $\varphi(X, Y)$ and additionally satisfies the following condition:

$$\forall X, Y. \psi(X, Y) \Rightarrow (|X| = \infty \Leftrightarrow |Y| = \infty),$$

where $|X| = \infty$ is a notation for “the set X is infinite”.

2 Assignment 2

In this assignment we work in “pureMSO” whose syntax (over some fixed alphabet A) is:

$$\psi ::= \neg\psi \mid \psi \vee \psi \mid \exists X. \psi \mid a(X) \mid X \subseteq Y \mid X \leq Y,$$

where $a \in A$, X ranges over set-variables, $a(X)$ means that all positions in X are labelled a , $X \leq Y$ means that all positions in X are $\leq$ all positions in Y , and other connectives and quantifiers are defined using De Morgan’s laws.

We say that a formula $\psi(X_1, \dots, X_n)$ is an (n, k) -formula if it has n free variables and it contains at most k nested quantifiers. More formally, a formula $\psi(X_1, \dots, X_n)$ is an (n, k) -formula for every $k \in \mathbb{N}$ if ψ has no quantifiers. If $\psi(X_1, \dots, X_{n+1})$ is an $(n+1, k)$ -formula then the formula $\psi'(X_1, \dots, X_n) = \exists X_{n+1}. \psi(X_1, \dots, X_n, X_{n+1})$ is an $(n, k+1)$ -formula. Negations and disjunctions of (n, k) -formulae are still (n, k) -formulae.

Fix an alphabet A and two natural numbers n, k . We consider “pureMSO” over finite words.

The goal is to show that there exists a specific deterministic automaton without final states, i.e. a tuple $\mathcal{A} = \langle A, Q, q_I \in Q, \delta: Q \times (A \times \{0, 1\}^n) \rightarrow Q \rangle$ with Q being a finite set. This $\mathcal{A}$ should have the property that for every (n, k) -formula $\psi(X_1, \dots, X_n)$ there should exist a subset $F_\psi \subseteq Q$ such that for every finite word w and every tuple $(Y_1, \dots, Y_n)$ of subsets of the domain of w the following conditions are equivalent:

1. the word w satisfies the formula $\psi(Y_1, \dots, Y_n)$,
2. the DFA $\langle A, Q, q_I, \delta, F_\psi \rangle$ accepts the word $(w, Y_1, \dots, Y_n)$ over the alphabet $A \times \{0, 1\}^n$.

Show that for every n, k , and A such $\mathcal{A} = \langle A, Q, q_I, \delta \rangle$ exists.

3 Assignment 3

Fix any finite alphabet A .

Take a sequence of ω -words $(w_n)_{n \in \mathbb{N}} \subseteq A^\omega$ and a single ω -word $w \in A^\omega$. We say that $\lim_{n \rightarrow \infty} w_n = w$ (the ω -word w is the *limit* of $(w_n)_{n \in \mathbb{N}}$) if for every $i \in \omega$ the sequence $(w_n(i))_{n \in \mathbb{N}}$ equals $w(i)$ for almost all $n \in \mathbb{N}$ (this is often called “pointwise convergence” — at each position $i \in \omega$ the i th letters of ω -words w_n stabilise and equal $w(i)$ for all except finitely many n).

Prove that the following conditions are equivalent for an ω -language $L \subseteq A^\omega$:

1. L is recognised by a non-deterministic Büchi automaton with all states accepting (formally $F = Q$). Such an automaton may have some transitions missing, i.e. for some $q \in Q$ and $a \in A$ there may be no $q' \in Q$ such that $(q, a, q') \in \delta$.
2. L is recognised by a deterministic Büchi automaton such that whenever $\delta(q, a) = q'$ and $q \notin F$ then also $q' \notin F$.
3. L is of the form $\{w \in A^\omega \mid \forall i \in \omega. w(0) \cdots w(i-1) \in K\}$ for some regular language $K \subseteq A^*$ — L is defined as the set of ω -words whose all finite prefixes belong to K .
4. L is closed wrt. pointwise convergence: if $(w_n)_{n \in \mathbb{N}} \subseteq L$ is a sequence of ω -words from L and $\lim_{n \rightarrow \infty} w_n = w$ is some ω -word $w \in A^\omega$ then also $w \in L$.